PEDIATRIC INTENSIVE CARE UNIT

Policy Name : Blood Sampling from Indwelling Catheter

Policy Number : BSFIC – PICU 3/3 (9)

	Full Name Title / Position	Signature	Date
Prepared by	MS. SIONY ESTIMA PICU Head Nurse		23.06.2024
Reviewed by	DR. SAMI TAHA Chief of PICU		23/06/2024
	DR. AHMED ALGHAMDI Chief of Pediatric Department		24/6/24
	MR. SAUD MOHD ZAHRANI Director of Nursing		23/6/24
	MR. MUTEB AL ESSA Director, Infection Control & Prevention		25/6/2024
	MS. AMAL KENANI Director, Quality and Patient Safety		26.06.2024
	DR. NASSER ASSIRI Medical Director		24/6/24
Approved by	DR. HASSAN UTHMAN AL-ZANDI Executive Director General		27/06/2024

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PURPOSE:

1. To avoid multiple blood extraction to patients having indwelling lines.
2. To maintain sterile technique and avoid infection during procedure.
3. To unify the procedure in taking samples from indwelling lines.

SCOPE:

First step: Blood investigations order by the physician.

Last step: Blood sample taken through aseptic technique.

ABBREVIATIONS:

ml - Milliliter

IPSG - International Patient Safety Goals

KVO - Keep vein open

PICC - Peripherally inserted central catheter

DEFINITIONS:

Central venous access device:

Central venous access device is a device designs and used for long term administration of medications and fluids into central veins.

MATERIAL / EQUIPMENT:

1. Appropriate blood sample tubes
2. Sterile 1/3/5/10 mL syringe
3. Chlorhexidine gluconate or 70% alcohol swab
4. Prefilled and labeled syringes with appropriate flush
5. Sterile gloves
6. Heparinized IV fluids
7. Appropriate request and barcode label to laboratory

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POLICY STATEMENT:

1. Hand hygiene is mandatory before and after accessing central lines.
2. Aseptic technique must be utilized for all aspects of central line care.
3. Competent nurse is responsible for drawing samples from central line / ports.
4. For PICC line, use only 1 and 3 ml syringes to avoid blockage of line.
5. If line is not use, always ask for fluids for KVO with heparin to avoid blocking the line.

PROCEDURE:

S. No	Procedure	Rationale
Preparatory phase:		
Obtain the writtrn order from physican		
1.	Identify the patient by using two identifiers (Name and medical record number – first identification)	Correct identification of patient as per IPSG goal 1
2.	Explain the procedure to the patient of parents.	To allay anxiety and maintain cooperation.
3.	Gather material and equipment for the procedure	For easy accessibility
Performance phase:		
4.	Hands hygien wear sterile gloves.	To maintain sterile technique.
5.	Assess all medications and infusions before selecting a port for sampling.	To select the right port for blood extraction
6.	Clamp the catheter and disconnect the infusion from the entering that is being sample.	Cap tubing that is disconnected using aseptic technique. If drawing from multi lumen central line, it is recommended to temporarily (more than 1 minute) stop all other infusions until blood drawing has been completed.
7.	Using vigorous friction for at least 15-30 seconds, prep the injection site with chlorhexidine gluconate or 70% alcohol swab.	To maintain sterile technique and not to infuse possible infection.

8.	Let it dry thoroughly.	Antiseptic maximum effectiveness is only achieved if antiseptic is allowed to air dry thoroughly.
9.	Luer lock the syringe to the injection site and unclamp the catheter.	Clamp triple lumen catheters with each syringe change.
10.	Confirm if indwelling line contains heparin.	Many lines are not heparin flushed secondary to heparin induced thrombocytopenia potential with frequency of blood draws and heparin flusher.
11.	If obtaining specimens from indwelling line that may contain heparin, using 10 mL syringe, first flush the line with 10 mL of normal saline. Remove the syringe and clamp the catheter.	If heparin is in the line it may affect laboratory results.
12.	Attach a new syringe and withdraw the first 5-10 mL of blood or 6 times the line volume (dead space volume of the catheter) and keep it to push back after sampling.	To make sure that infusions are not taken as samples and to have an accurate results.
13.	With a new syringe, withdraw volume of blood equal to the total amount of blood needed for all blood samples.	
14.	If blood culture is being taken, use a separate syringe for the blood culture and a different syringe for other laboratory.	To avoid contamination.
15.	Push back slowly the blood aspirated prior to taking sample.	
16.	Using another 10 mL syringe flush the catheter with normal saline.	
17.	Reconnect the infusion and start according to previous setting. If no infusion flush with heparinized normal	To avoid blockage of central lines.

	saline as order.	
18.	Throw all disposable items used according to infection control policy.	
19.	Check the infusion site and document the whole procedure to nurse's flow chart including amount of blood taken and normal saline infused.	To maintain intake and output record.
	Follow up phase:	
20.	Discard all disposable items into appropriate waste bags	Proper disposal of waste materials according to infection control policy.
21.	Check the infusion site for any bleeding from port and make sure that infusions are going properly	To ensure the patency of line and right infusions are ongoing via ports at correct rate
22.	Document the whole procedure to nurses flow chart and nurse's note including: 22.1. Amount of blood taken 22.2. Amount of normal saline infused 22.3. Strength of back flow 22.4. Blood investigations 22.5. Adverse effects 22.6. Patient's reaction	Documentation ensures accountability and continuity of care and also to maintain legal record

DOCUMENTATION

1. OASIS system
2. PICU flow chart (FM 052)

RESPONSIBILITIES:

Physician is responsible to order needed blood investigations and if can be withdraw from indwelling lines.

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Head Nurse has to make sure that all staff completed their competency before they can do the procedure.

Charge Nurses are to supervise bedside nurse and make sure that procedure is done aseptically.

Staff Nurses have to maintain sterile technique throughput the procedure and document afterwards.

MEASUREMENT:

1. 100% of staff must have completed competency check-off before doing the procedure.
2. 100% of blood sampling from indwelling lines shall be done under aseptic technique.

REFERENCES:

1. Textbook of Medical Surgical Nursing Brunner & Suddarth's 14th edition.
2. Lippincott Manual of Nursing Practice 11th edition.

CROSS REFERENCES:

1. Infection Control Policy

DISTRIBUTION:

1. Nursing Administration
2. Quality Department
3. Office of the Executive Director General

REVIEW SUMMARY

1. Additional abbreviations / Definitions
2. Addition of Policy Statement

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